

The differential $p(\mathbf{x}) d\mathbf{x}$ can be sampled over by MCMC, knowing only $\pi(\mathbf{x})$; no problem. The $\pi(\mathbf{x})$ in the denominator of the integrand can also be readily computed. But we have, in general, no easy way to calculate that pesky normalizing constant, $\int \pi(\mathbf{x}) d\mathbf{x}$.

Sometimes, though not often, you can construct a function $\pi(\mathbf{x})$ that both resembles f and also can be integrated analytically, so that the normalizing constant is knowable. Then, yes, by all means use MCMC to sample $\pi(\mathbf{x})$. In this case the idea of recording only every m th step, after choosing m large enough so that the points thus chosen are independent samples, is not a bad idea after all. In fact you'll have to do this if you expect to use the error estimate in equation (7.9.3) as written.

Finally, if the integral that you really want is

$$J \equiv \frac{\int f(\mathbf{x})\pi(\mathbf{x}) d\mathbf{x}}{\int \pi(\mathbf{x}) d\mathbf{x}} = \int f(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} \quad (15.8.17)$$

with $f(\mathbf{x})$ and $\pi(\mathbf{x})$ both known (and $p(\mathbf{x})$ only implied), then MCMC is exactly what you need. It provides uniform samples over $p(\mathbf{x})d\mathbf{x}$, and no calculation of a normalizing constant is needed.

CITED REFERENCES AND FURTHER READING:

- Hastings, W.K. 1970, "Monte Carlo Sampling Methods Using Markov Chains and Their Applications," *Biometrika*, vol. 57, pp. 97–109.[1]
- Gilks, W.R., Richardson, S., and Spiegelhalter, D.J., eds. 1996, *Markov Chain Monte Carlo in Practice* (Boca Raton, FL: Chapman & Hall/CRC), especially Chapter 1.[2]
- Gamerman, D. 1997, *Markov Chain Monte Carlo: Stochastic Simulation for Bayesian Inference* (London: Chapman & Hall). [3]
- Neal, R.M. 1993, "Probabilistic Inference Using Markov Chain Monte Carlo Methods," *Technical Report CRG-TR-93-1*, Department of Computer Science, University of Toronto. Available at <http://www.cs.toronto.edu/~radford/ftp/review.pdf>. [4]
- Casella, G., and George, E.I. 1992, "Explaining the Gibbs Sampler," *American Statistician*, vol. 46, no. 3, pp. 167–174.[5]
- Tanner, M.A. 2005, *Tools for Statistical Inference: Methods for the Exploration of Posterior Distributions and Likelihood Functions*, 3rd ed. (New York: Springer).
- Liu, J.S. 2002, *Monte Carlo Strategies in Scientific Computing* (New York: Springer).
- Beichl, I., and Sullivan, F. (eds.) 2006, *Computing in Science and Engineering*, special issue on Monte Carlo Methods, vol. 8, no. 2 (March/April), pp. 7–47.

15.9 Gaussian Process Regression

Some types of statistical models do not depend on knowing (or guessing) parameterized functional forms, and thus lie outside of the parameter-fitting paradigm that has thus far occupied our attention. As an alternative to assuming that our data have some functional form, we can assume that they have some statistical property. A common example is to assume that the data, viewed as an entire set, is drawn from some multivariate normal (Gaussian) distribution in a high-dimensional space. That distribution is allowed to have a complicated correlation structure: The individual data points are *not* assumed to be independent. We can then ask, *given* the

data points that we observe, what are the most probable values for other quantities of interest, for example the values of variables at points other than the ones measured. Of course, as previously, we are also encouraged to ask not just about the most probable values, but about the whole distribution around the most probable values. This general scheme is called *Gaussian process regression*.

We have already met examples of Gaussian process regression twice before in this book, though under different names. In §3.7 we discussed *kriging* as a multidimensional interpolation technique. Later, in §13.6, we discussed *linear prediction*, mostly in the context of one-dimensional data such as time series. Here we can usefully merge some of the ideas in those two sections.

As we presented it in §3.7, kriging was an interpolation, not a fitting, technique. This was evident from the facts that (i) the interpolated function output by the `Krig` object went exactly through the measured data points, and (ii) we never discussed how to input measurement errors. However, the `Krig` object's constructor did have an argument `err`, introduced with the mysterious remark that you should leave it set to `NULL` until you read §15.9. Well, here we are!

We did incorporate measurement errors in §13.6, although they were there called *noise*. In particular, equations (13.6.6) and (13.6.7) can be used (after some change of notation and algebraic manipulation) to derive the appropriate generalization of equations (3.7.14) and (3.7.15) to the case where the measurements y_i , $i = 0, \dots, N - 1$, have errors characterized by some covariance matrix Σ . In most cases Σ will be simply a diagonal matrix with elements σ_i^2 , the squares of the individual errors. The answers are

$$\hat{y}_* = \mathbf{V}_* \cdot (\mathbf{V} - \Sigma')^{-1} \cdot \mathbf{Y} \quad (15.9.1)$$

and

$$\text{Var}(\hat{y}_*) = \mathbf{V}_* \cdot (\mathbf{V} - \Sigma')^{-1} \cdot \mathbf{V}_* \quad (15.9.2)$$

where

$$\Sigma' \equiv \begin{pmatrix} \Sigma & 0 \\ 0 & 0 \end{pmatrix} \quad (15.9.3)$$

That is, we simply subtract Σ (suitably augmented by bordering zeros) from $\mathbf{V}$ (equation 3.7.13) before inverting the matrix. The argument `err`, input as the σ_i 's (not squared), does this for the case of diagonal measurement errors. Note that `err` has type `Doub*`. If your errors are stored in a `VecDoub`, then you'll send `&err[0]` to the `Krig` constructor. (Sorry about this hack. The purpose was to make `NULL` a possible default value.)

So, no new code is needed in this section. In `Krig`, you already have a serviceable multidimensional Gaussian process regression fitting routine, all ready to go.

When you are fitting, rather than interpolating, it is a good idea to pay more attention to the choice of variogram model than we did in §3.7. While for simple applications there is nothing wrong with the power-law model implemented in the `Powvargram` object

$$v(r) = \alpha r^\beta \quad (15.9.4)$$

several other models are widely used. These include the *exponential model*,

$$v(r) = b[1 - \exp(-r/a)] \quad (15.9.5)$$

the *spherical model*,

$$v(r) = \begin{cases} b \left(\frac{3}{2} \frac{r}{a} - \frac{1}{2} \frac{r^3}{a^3} \right) & 0 \leq r \leq a \\ b & a \leq r \end{cases} \quad (15.9.6)$$

and various anisotropic models for which $v(\mathbf{r})$ is not just a function of the magnitude r . See [1,2] for derivations and examples.

We should also mention the so-called *nugget effect*, though, in our opinion, its name vastly outshines its utility. If $v(\mathbf{r})$ does not go to zero as $\mathbf{r} \rightarrow 0$, but instead goes to some constant v_0 , then the resulting variogram describes a distribution that decorrelates by some finite amount in an infinitesimal distance. That is, if you find a gold nugget at location $\mathbf{x}$, there is no certainty that you'll find another one at location $\mathbf{x} + \delta\mathbf{x}$, no matter how small you make $\delta\mathbf{x}$. Some practitioners deem it desirable to allow for a nonzero nugget effect, allowing nonzero values of v_0 when they empirically fit $v(r)$ from a data set. That seems debatable to us; but in deference to such opinion we have given the *Powvargram* constructor an otherwise undocumented argument, *nug*, for feeding in the value v_0 of your choice. (We draw the line at actually fitting for such a parameter!)

Beyond debatable, and actually incorrect, however, is to confuse the nugget effect with the effect of measurement error. They seem superficially similar: Measurement error also decorrelates measured values, even at arbitrarily small distances (even zero). Conceptually, and mathematically, however, they are different. Referring to equation (3.7.13), a nugget effect adds a constant positive value to all the *off-diagonal* v_{ij} 's. Measurement errors, on the other hand, subtract (not necessarily constant) negative values from the *diagonal* v_{ii} 's. These actions do not have equivalent effects on equations (3.7.14) and (3.7.15). This can readily be seen in Figure 15.9.1, which may also help elucidate the difference between kriging interpolation and kriging fitting. Only panel (d) in the figure shows a correct use of kriging for data with measurement errors, that is, kriging fitting with errors σ_i . Panels (b) and (c) show the results of kriging interpolation with and without a nugget effect. One sees that even with a positive nugget, the interpolated curve goes exactly through the data points, which is incorrect when measurement errors are significant. The legitimate use of kriging interpolation (as in §3.7) is for smooth functions that are “exactly” known at scattered points. Kriging fits using σ_i 's (this section) are for data with errors.

CITED REFERENCES AND FURTHER READING:

- Cressie, N. 1991, *Statistics for Spatial Data* (New York: Wiley).[1]
 Wackernagel, H. 1998, *Multivariate Geostatistics*, 2nd ed. (Berlin: Springer).[2]
 Isaaks, E.H., and Srivastava, R.M. 1989, *Applied Geostatistics* (New York: Oxford University Press).
 Rasmussen, C.E., and Williams, C.K.I. 2006, *Gaussian Processes for Machine Learning* (Cambridge, MA: MIT Press).
 Rybicki, G.B., and Press, W.H. 1992, “Interpolation, Realization, and Reconstruction of Noisy, Irregularly Sampled Data,” *Astrophys. J.*, vol. 398, pp. 169–176.
 Deutsch, C.V., and Journel, A.G. 1992, *GSLIB: Geostatistical Software Library and User's Guide* (New York: Oxford University Press).

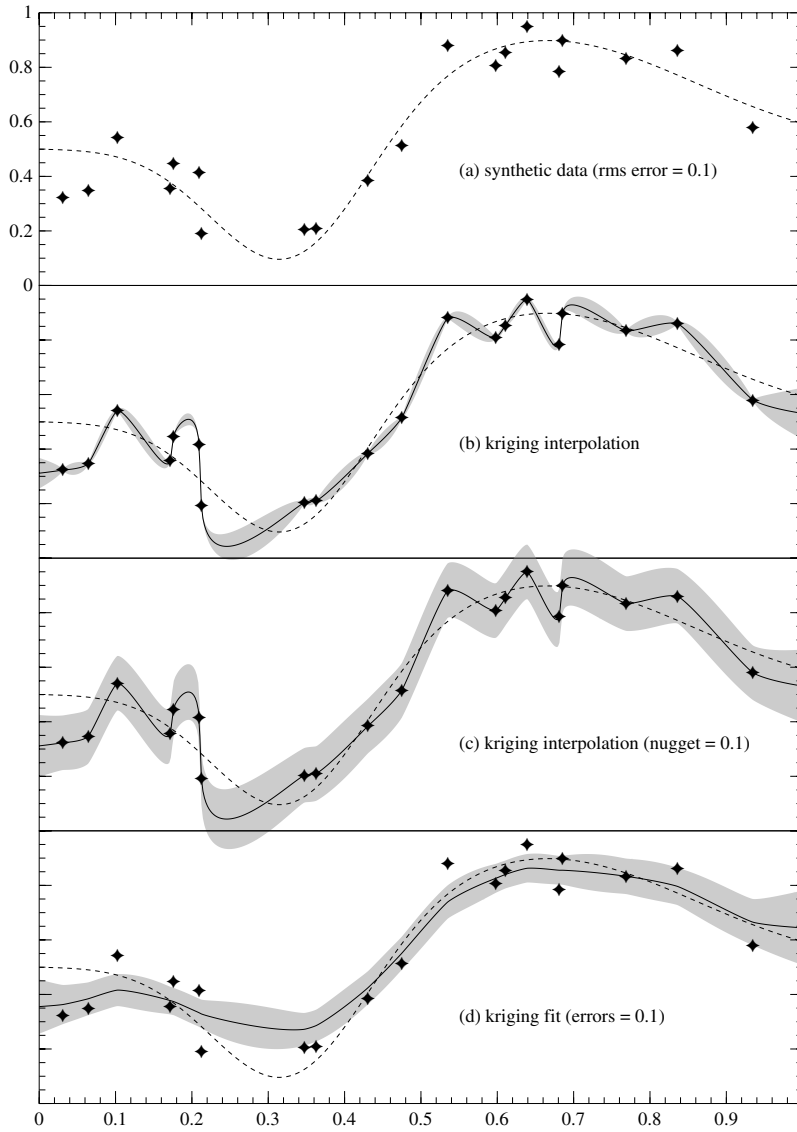


Figure 15.9.1. One-dimensional examples of interpolation and fitting by kriging. (a) Synthetic data points generated from the known curve (dashed line) with Gaussian errors (r.m.s. magnitude 0.1). (b) Result of kriging interpolation. Equation (3.7.14) is plotted as the solid line, while the 1- σ estimated interpolation errors (3.7.15) are shown as the shaded band. The interpolation error is seen to be meaningless for data with measurement errors. (c) Same as (b), but with a nugget effect of 0.1. (d) Result of kriging fit (equations 15.9.1 and 15.9.2) using the actual measurement errors. This is the correct use of kriging for data with errors.